

5.3 Introduction to Boundary Value Problem

```
In [1]: using CairoMakie
using LinearAlgebra
using LinearSolve
using SparseArrays
using BenchmarkTools
```

In this notebook, we are going to find the temperature distribution of a uniformly heated plate. From physics, the governing equation can be written as

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) = -q(x)$$

We usually assume that k is a constant so the above equation can be simplified to be

$$k \frac{d^2 T}{dx^2} = -q(x)$$

For uniform heating, $q(x) = 1$. Assume that $x \in [0, 1]$ with the boundary conditions $T(x = 0) = 5$ and $dT/dx(x = 1) = 0$. We want to find how T changes with x . Assume that $k = 0.1$

Using the finite difference approximation

$$\frac{d^2 T}{dx^2} \approx \frac{T_{i-1} - 2T_i + T_{i+1}}{\Delta^2}$$

So at every point in the domain, we will have

$$k \frac{d^2 T}{dx^2} \approx \frac{T_{i-1} - 2T_i + T_{i+1}}{\Delta^2} = -1$$

We will now split our domain into 5 sub intervals and apply the equation above onto each of the interior points

```
In [2]: Delta=0.2
x=0:Delta:1
α=5.0
ThermalConductivity=0.1
```

0.1

$$k \frac{d^2 T}{dx^2}(x_i = x_2) \approx \frac{T_1 - 2T_2 + T_3}{\Delta^2} = -1$$

$$k \frac{d^2 T}{dx^2}(x_i = x_3) \approx \frac{T_2 - 2T_3 + T_4}{\Delta^2} = -1$$

$$k \frac{d^2 T}{dx^2}(x_i = x_4) \approx \frac{T_3 - 2T_4 + T_5}{\Delta^2} = -1$$

$$k \frac{d^2 T}{dx^2}(x_i = x_5) \approx \frac{T_4 - 2T_5 + T_6}{\Delta^2} = -1$$

We also know that at $T(x = 0) = 5$, so $T_1 = 5$.

At $x = 1$, we are given that $(T_6 - T_5)/\Delta = 0$, so $T_6 - T_5 = 0$

We can put this in matrix form as

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1/\Delta^2 & -2/\Delta^2 & 1/\Delta^2 & 0 & 0 & 0 \\ 0 & 1/\Delta^2 & -2/\Delta^2 & 1/\Delta^2 & 0 & 0 \\ 0 & 0 & 1/\Delta^2 & -2/\Delta^2 & 1/\Delta^2 & 0 \\ 0 & 0 & 0 & 1/\Delta^2 & -2/\Delta^2 & 1/\Delta^2 \\ 0 & 0 & 0 & 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \end{Bmatrix} = \begin{Bmatrix} 5 \\ -1/k \\ -1/k \\ -1/k \\ -1/k \\ 0 \end{Bmatrix}$$

If we solve the equation above, we can get the approximate values of the temperature T_i

```
In [3]: n=length(x)
d=zeros(n)
dl=zeros(n-1)
du=zeros(n-1)
C=zeros(n)
```

6-element Vector{Float64}:

```
0.0
0.0
0.0
0.0
0.0
0.0
```

The code below constructs the matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1/\Delta^2 & -2/\Delta^2 & 1/\Delta^2 & 0 & 0 & 0 \\ 0 & 1/\Delta^2 & -2/\Delta^2 & 1/\Delta^2 & 0 & 0 \\ 0 & 0 & 1/\Delta^2 & -2/\Delta^2 & 1/\Delta^2 & 0 \\ 0 & 0 & 0 & 1/\Delta^2 & -2/\Delta^2 & 1/\Delta^2 \\ 0 & 0 & 0 & 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \end{Bmatrix} = \begin{Bmatrix} 5 \\ -1/k \\ -1/k \\ -1/k \\ -1/k \\ 0 \end{Bmatrix}$$

```
In [4]: du[1]=0.0
d[1]=1.0
C[1]=5.0
for i=2:n-1
    dl[i-1]=(1.0/Delta^2)
    d[i]=-(2.0/Delta^2)
    du[i]=(1.0/Delta^2)
    C[i]=-1.0/ThermalConductivity
end
dl[n-1]=-1
d[n]=1.0
C[n]=0.0

# Construct the specialized structured matrix
A = Tridiagonal(dl, d, du)
```

```
6×6 Tridiagonal{Float64, Vector{Float64}}:
 1.0  0.0  .  .  .  .
25.0 -50.0 25.0 .  .  .
 .  25.0 -50.0 25.0 .  .
 .  .  25.0 -50.0 25.0 .
 .  .  .  25.0 -50.0 25.0
 .  .  .  . -1.0 1.0
```

```
In [5]: A_sparse = sparse(A)
```

```
6×6 SparseMatrixCSC{Float64, Int64} with 16 stored entries:
 1.0  0.0  .  .  .  .
25.0 -50.0 25.0 .  .  .
 .  25.0 -50.0 25.0 .  .
 .  .  25.0 -50.0 25.0 .
 .  .  .  25.0 -50.0 25.0
 .  .  .  . -1.0 1.0
```

```
In [6]: prob=LinearProblem(A,C)
#@btime y=LinearSolve.solve(prob)
@time Tuniform=LinearSolve.solve(prob)
```

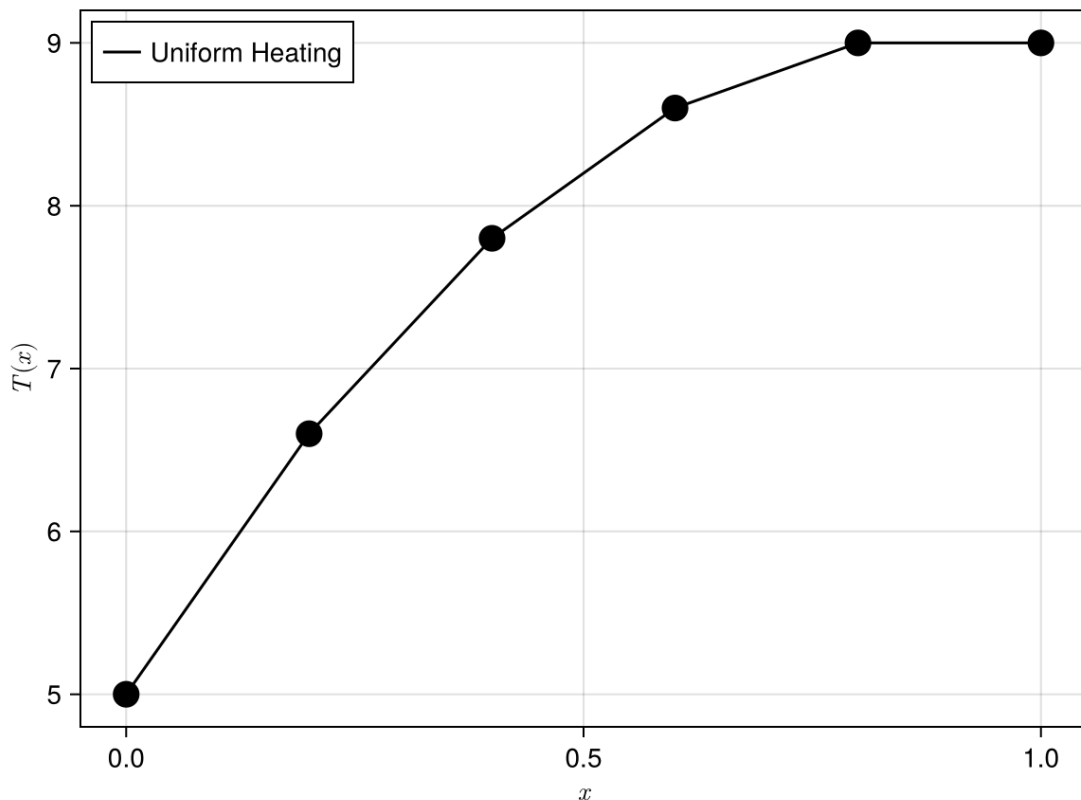
```
3.034487 seconds (18.40 M allocations: 942.663 MiB, 12.16% gc time, 100.
00% compilation time)
retcode: Success
u: 6-element Vector{Float64}:
 5.000000000000001
 6.6000000000000005
 7.800000000000001
 8.600000000000001
 9.000000000000002
 9.000000000000002
```

```
In [7]: probSparse=LinearProblem(A_sparse,C)
@time y=LinearSolve.solve(probSparse, UMFPACKFactorization())
```

```
0.104771 seconds (388.91 k allocations: 20.009 MiB, 99.20% compilation t
ime)
retcode: Success
u: 6-element Vector{Float64}:
 5.0
 6.6000000000000005
 7.800000000000001
 8.600000000000001
 9.000000000000002
 9.000000000000002
```

Note that tridag solver is faster since it is specific for tridiagonal systems. Sparse solver is for more general sparse matrices, hence it is slower than tridag solver.

```
In [15]: xplot=0:0.01:1
fig = Figure()
ax = Axis(fig[1, 1], xlabel = L"x", ylabel = L"T(x)")
lines!(ax,x,Tuniform,color=:black,label="Uniform Heating")
scatter!(ax,x,Tuniform,color=:black,markersize=20)
axislegend(ax,position=:lt)
save("../figures/05p01IntroBVP01.svg",fig)
fig
```



In [9]: `y[end]`

9.000000000000002

In this next example, the plate is heated but not uniformly. The heating increases with x and can be modelled as

$$q(x) = (1 + 2x)$$

The governing equation is

$$k \frac{\partial^2 T}{\partial x^2} = -q(x)$$

We can go through the same steps above to get

The code below constructs the matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1/\Delta^2 & -2/\Delta^2 & 1/\Delta^2 & 0 & 0 & 0 \\ 0 & 1/\Delta^2 & -2/\Delta^2 & 1/\Delta^2 & 0 & 0 \\ 0 & 0 & 1/\Delta^2 & -2/\Delta^2 & 1/\Delta^2 & 0 \\ 0 & 0 & 0 & 1/\Delta^2 & -2/\Delta^2 & 1/\Delta^2 \\ 0 & 0 & 0 & 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \end{Bmatrix} = \begin{Bmatrix} 5 \\ -q(x_2)/k \\ -q(x_3)/k \\ -q(x_4)/k \\ -q(x_5)/k \\ 0 \end{Bmatrix}$$

```
In [10]: for i=2:n-1
           C[i]=-(1.0+2*x[i])/ThermalConductivity
         end
```

```
In [11]: prob=LinearProblem(A,C)
#@btime y=LinearSolve.solve(prob)
@time Tlinear=LinearSolve.solve(prob)
```

```
0.000009 seconds (20 allocations: 1.188 KiB)
retcode: Success
u: 6-element Vector{Float64}:
 4.999999999999992
 8.199999999999994
10.839999999999996
12.76
13.800000000000002
13.800000000000004
```

```
In [12]: xplot=0:0.01:1
fig = Figure()
ax = Axis(fig[1, 1], xlabel = L"x", ylabel = L"T(x)")
lines!(ax,x,Tuniform,color=:black,label="Uniform Heating")
lines!(ax,x,Tlinear,color=:blue,label="Linear Heating")
axislegend(ax,position=:lt)
fig
```

